Health management in Asian aquaculture

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This document contains the technical papers presented at the Regional Expert Consultation on Aquaculture Health Management in Asia and the Pacific, jointly organized by the Fish Health Section of the Asian Fisheries Society (FHS/AFS) and the Fishery Resources Division (FIR) of the Fisheries Department of FAO, held at Universiti Pertanian Malaysia, Serdang, Selangor, Malaysia, 22–24 May 1995. The document also contains the recommendations made at the Expert Consultation. The full report of the Expert Consultation is published in the *FAO Fisheries Report* No. 529 (FAO, Rome, 1995, 24 p.).

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ABSTRACT

In 1994, world aquaculture production reached 25.5 million mt, valued at US$ 39.83 billion. Asia contributed 89.9% of this total, and has since continued to dominate global production. The drive to produce more fish and shellfish to meet the growing demand has lead many aquaculturists in Asia to intensify their operations. In many instances, the complex balance between the fish/shellfish and the environment is not well understood, the organism under culture subsequently becoming stressed and prone to infections. As we have already witnessed, disease has been and will continue to be a major constraint to the development of the aquaculture industry. Considering the FAO's priority on developing sustainable aquaculture, the large Asian contribution to global aquaculture production and the seemingly high losses of revenue due to diseases and health-related problems, FAO, in consultation with the Network of Aquaculture Centres in the Asia-Pacific (NACA), the Aquatic Animal Health Research Institute (AAHRI), the South East Asian Fisheries Development Centre (SEAFDEC) and the Universiti Pertanian Malaysia (UPM), and in collaboration with the Fish Health Section of the Asian Fisheries Society (FHS/AFS), organized a Regional Expert Consultation on Aquaculture Health Management in Asia and the Pacific, which was held at the Universiti Pertanian Malaysia in Serdang, Malaysia in May 1995. This document comprises the technical papers presented

at the Consultation, and is a supplement to the report of the consultation, *FAO Fisheries Report* No. 529 (FAO, Rome, 1995. 24 p.)

(Key words: Asia, Pacific, Aquaculture, Fish disease, Health management, Quarantine)

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CONCLUSIONS AND RECOMMENDATIONS OF THE REGIONAL EXPERT CONSULTATION ON AQUACULTURE HEALTH MANAGEMENT IN ASIA AND THE PACIFIC, HELD AT THE UNIVERSITI PERTANIAN MALAYSIA IN SERDANG, MALAYSIA IN MAY 1995 (*FAO Fisheries Report* No. 529, Rome, FAO. 1995. 24p.)

4.1 *General conclusions and recommendations*

4.1.1 *In light of the growing importance of aquaculture to global food supply and economic well-being and the highly significant contribution made by the Asian region to the world aquaculture production, the consultation* ***recognized*** *that it is imperative to ensure that the Asian aquaculture sector is developed in a sustainable manner.*

4.1.2. *Considering the past experiences with disease problems and the related economic losses to Asian aquaculture, the consultation* ***agreed*** *that diseases and related health problems are one of the most serious deterrents to the sustainable development of aquaculture in Asia and the Pacific.*

4.1.3. *It was* ***recognized*** *that not only health-related problems have a direct bearing on the aquatic environmental quality, but also that the future of the entire aquaculture industry will depend on the continuing availability of water and other resources with adequate quality.*

4.1.4. *The consultation* ***agreed*** *that there is an urgent need for an effective regional health management program and* ***suggested*** *three thrust areas for such a program; i) Research, diagnosis and information, ii) Training and extension, and iii) Quarantine and legislation.*

4.2. *Conclusions and recommendations on research, diagnosis and information*

Aquatic animal health research needs for a sustainable aquaculture industry in Asia have been recently reviewed at the Fish Health Management in Asia- Pacific Workshop organized by the ADB/NACA in Pusan, South Korea in October 1990 and at the workshop organized by the FHS/AFS and SIFR during the Second Symposium on Diseases in Asian Aquaculture held in Phuket, Thailand in October 1993. The consultation **endorsed** the specific recommendations made at the workshops and **emphasized** that these recommendations should be considered when formulating future aquatic animal health research programs for Asia. The consultation recognized the importance of regional cooperation in aquatic animal health research, diagnosis and information, and the following recommendations were made after carefully considering the past activities, present status, future directions, and the institutional and manpower capabilities in the region.

4.2.1. The Asian marine shrimp culture industry has expanded dramatically over the past decade. Although this growth is generated by market demand and short- term gains, and it represents only a small proportion of global aquaculture production, the contribution of this sector to the GNP of developing countries in the region is substantial. Recently, there have been a number of severe outbreaks of disease in Asian shrimp culture, mainly due to viral pathogens, which have caused serious economic losses to the regional shrimp industry. Considering the above:

*The consultation **reiterated** that viral diseases of shrimp were an extremely critical issue, more so than viral diseases of fish at present, and that immediate efforts must be made to develop research programs to combat viral disease problems in the Asian shrimp industry. It also **recognized** that medium- and long-term efforts should seek to develop the capacity for research on fish viruses, as such disease problems would likely occur in the future.*

4.2.2. In Asia, most countries do not appear to have adequate infrastructure for disease diagnosis and research on aquaculture health. Among the countries in the region, there are differences in prioritized species and systems for culture. The specific emphasis given by the governments in the region for establishing national aquatic health facilities appears to vary with their country's economic status. Therefore:

*The consultation **realized** that the primary responsibility for establishment and operation of national aquaculture health facilities, including diagnostic and research laboratories, remains with the national governments. Regional and international organizations should be encouraged to assist governments in strengthening such facilities*.

4.2.3. The disease diagnosis and research capabilities found in some countries in Asia and the Pacific are quite advanced. Trained diagnosticians, advanced laboratory facilities, and excellent aquaculture health centers are available and are functioning. Considering the fact that national-level capacity building in aquaculture health research and diagnosis is a long-term objective to achieve:

*The consultation **strongly emphasized** the importance of strengthening Asia's regional centers in aquaculture health management as a short-term activity. The participation of such centers in supporting capacity building in other countries of the region, through regional cooperation strategies, should be **strongly encouraged**. The consultation **agreed** that regional cooperative efforts towards strengthening aquaculture health centers could be further reinforced through increased international cooperation*.

4.2.4. Regional cooperation is encouraged to address aquaculture health problems in Asia. However, an effective regional networking mechanism with respect to aquaculture health management, which could bring effective regional cooperation, does not exist. With the expansion of regional aquaculture and the more frequent emergence of disease problems, a good networking mechanism could be very effective and would strengthen the regional aquaculture health management mechanism. Therefore:

*The importance of regional networking was **emphasized** as a cost-effective means of strengthening the national capacities for aquaculture health management. Increased efforts are required to promote the exchange of information, training, technical assistance and expertise to enhance such regional cooperative actions. These cooperative exchanges should include the more technically advanced countries of the region.*

4.2.5. In most Pacific countries, aquaculture is still in its infancy. However, the Pacific countries are interested in developing aquaculture and some have already embarked upon comprehensive programs. It is obvious that with the development of aquaculture in those countries, disease and health problems will emerge as major deterrents to the industry, if preventive and mitigating measures are not taken well in advance. Thus:

*The consultation **emphasized** that the Pacific countries could learn from experiences in the development of aquaculture in Asian countries*.

4.2.6. There are reports from the region about human zoonoses related to aquaculture. Some institutions are already addressing such problems, in particular the epidemiological aspects.

*The consultation **discussed** the possible merits of research on human health aspects of aquatic animal disease and **recognized** its importance to human well-being. However, it was noted that this subject was broader than aquaculture health management, and the consultation **recommended** that it be left to other initiatives*.

4.2.7. Genetic manipulation and the development of disease/pathogen resistant strains have been attempted with many fish and livestock species, but with limited success. The possible benefits of work on the genetic aspects of disease were discussed, particularly the possibility of incorporating disease resistance through selective breeding programs. Experience in the animal husbandry field suggested that such an approach was not promising. Thus,

*The consultation **recommended** that at this point in time research emphasis should not be given to the genetic aspects of disease.*

4.2.8. Effective dissemination of research findings, both regionally and internationally, is extremely vital to the advancement of scientific research. Many scientists in Asia appear to publish their research findings in local journals and reports, written in national languages. Although this grey literature has limited accessibility and use outside their country, it could provide vital information. Nevertheless, publishing research findings in peer- reviewed international journals will undoubtedly strengthen regional research.

*The consultation **endorsed** the importance of publishing fish health research findings in peer-reviewed journals and recommended that efforts should be made to **enhance** the availability and exchange of grey literature.*

4.2.9. The Asian Fish Health Reference Centre of the Fish Health Section of the Asian Fisheries Society has been established at the Universiti Pertanian Malaysia (UPM), where the bibliographic information pertaining to regional fish health topics is deposited. UPM has a fully fledged library with modern facilities and is willing to provide assistance to FHS/AFS to further strengthen its reference center.

*The consultation **strongly recommended** that participating countries supply aquaculture health literature to UPM, and that UPM should be encouraged to collate and disseminate such information to the countries of the region*.

4.2.10. Information technology in Asia, especially in many South East Asian countries, is sound, and many institutions are now capable of using it on a routine basis. These modern information technologies could be effectively utilized to help competent dissemination of fish health literature within and outside the region. Considering the availability of such advanced facilities in some countries in the region:

*The consultation **recommended** that the use of e-mail, bulletin boards and advanced information exchange technologies should be further explored as an effective means to exchange information on aquaculture health issues.*

4.3. *Conclusions and recommendations on training and extension*

4.3.1. In most Asian universities, fish health is not taught within their fisheries, aquaculture, animal husbandry or veterinary curricula. There have been some attempts to incorporate fish health into university curricula, with little success. Considering the urgent need for trained manpower in the field of fish health to support sustainable development of regional aquaculture:

The consultation suggested that universities should include aquatic animal health management components in curricula for fisheries, aquaculture and/or veterinary programs, but it was noted that very few appropriately skilled instructors are currently available. It was therefore suggested that, in the short term and until such expertise is more generally available, regional centers of excellence should provide the necessary training.

4.3.2. There are many directories of experts in fisheries and aquaculture available in Asia and the Pacific. However, a concise directory on fish health scientists, providing their specialities and services, has yet to be produced. Such a directory would be of tremendous value in strengthening aquaculture health management in Asia and the Pacific. Considering the above:

The consultation recommended that a directory of scientists in the region with expertise in aquaculture health should be produced. The listing should include specific information regarding each expert's skills.

4.3.3. Training in fish health has been undertaken by a number of regional institutions. SEAFDEC (Philippines) and AAHRI (Thailand) are two institutions which conduct specific training programs for regional aquaculturists and extensionists on a regular basis. These institutions offer courses at different levels with different approaches addressing the specific requirements and needs of trainees and their countries. However, there has not been an attempt to compare and contrast the curricula program. Considering the diversity of technical training courses offered in the region:

*The consultation suggested that attempts be made to combine the best features of the integrated health management approach (as in AAHRI programs) with the best features of the specialist discipline approach (as in SEAFDEC programs). Such integration could deliver more uniform training programs. It was **further recommended** that standardized currricula should be developed to assist national and regional centers in the development of training courses. It was **suggested** that training courses for trainers could be conducted in regional centers of excellence and or more advanced countries such as Japan, UK, USA or Australia.*

4.3.4. Over the years, with the help of donor agencies, various Asian institutions have provided fish health training to a considerable number of regional aquaculturists and extensionists. However, it has been repeatedly highlighted that one of the major constraints to developing an effective health management strategy for regional countries is lack of trained personnel with practical experience. This raises the question as to the effectiveness of training programs. Considering the above:

*The consultation **recommended** that efforts should be made to evaluate the effectiveness of training courses and that such evaluation should be conducted by the training institution using independent evaluators.*

4.3.5. With respect to the type of training offered, field-based training for field personnel and laboratory-based training for laboratory workers was seen as important. However:

*The consultation also **recognized** that laboratory-based diagnosticians and researchers should be familiar with on-farm conditions and field-based extensionists should be familiar with diagnostics and therapy so that informed decisions on control and treatment can be made.*

4.3.6. Considering the severity of the recent disease outbreaks in Asia and their spread pattern:

The consultation recognized the importance of epidemiology/epizootiology in providing solutions to aquaculture health problems, and noted the dearth of epidemiologists/epizootiologists currently working in the field. It was recommended that immediate efforts be made to rectify this situation.

4.4. *Conclusions and recommendations on quarantine and legislation*

4.4.1. There is strong evidence that many of the disease outbreaks which have occurred recently in Asian aquaculture are linked to the introduction of new pathogens through trans-boundary movement of aquatic species. The distributions of many shrimp viral diseases and of epizootic ulcerative syndrome (EUS) appear to have direct relationships with the movement of species. As aquaculture grew over the past two decades, trans-boundary movement of aquatic organisms became intense, creating the importance and urgency of establishing quarantine systems, protocols and guidelines for the movement of aquatic species. Together with efforts to implement stringent consumer protection strategies and to ensure sustainable development of global aquaculture, guidelines, protocols and codes of practice for aquatic animal health quarantine and movement have been developed by various agencies and institutions. After careful consideration of the recent global and regional developments with respect to aquatic animal health and quarantine:

*The consultation **strongly emphasized** the importance of documenting economic justifications and cost-benefit analyses for introduction of legislation and associated quarantine systems. Policy makers are only likely to be convinced by strong economic arguments. Marketing and trading issues are also important concerns in relation to the overall transfer of aquatic animals and pathogens from one place to another.*

4.4.2. In order to develop practical and realistic guidelines and legislation on quarantine and health, it is imperative to investigate carefully the regional fish health status, including current trends in the movement of species, occurrence and pattern of spread of diseases, causative agents and their origin, and existing guidelines and legislation. It is perceived that the required literature is available in Asia and that this information needs to be compiled into an effective format. Thus:

*The consultation strongly recommended that a literature search be conducted, as soon as possible, to develop a preliminary list of pathogens of aquatic animals in the countries of the region. The list would help in the assessment of quarantine and aquatic animal health legislation. However, it was **recognized** that further research on the epizootiology and epidemiology of aquatic animal diseases will be required to develop a comprehensive list.*

4.4.3 In an effort to facilitate international trade in aquatic animals and animal products, the Office Internationale des Epizooties (OIE) will soon publish the International Aquatic Animal Health Code and Diagnostic Manual, the prototype of which is now being reviewed. The Code and Manual attempts to achieve this aim by providing detailed definitions of minimum health guarantees, based on standard methods for laboratory examinations, to be required for trading partners in order to avoid the risk of spreading aquatic animal diseases.

*The consultation **discussed** the OIE Aquatic Animal Health Code and Diagnostic Manual. It was **emphasized** that the Code and Manual are based on temperate species and aquaculture systems in developed countries, and that care should be taken in applying these guidelines to the Asian region. The consultation **stressed** the importance of evaluation of the OIE Code and Manual in relation to the situation in the Asian region, and their further development to address the species and systems of Asia.*

4.4.4. Over the years, certain South East Asian countries such as Indonesia and Malaysia have attempted to introduce aquatic animal quarantine procedures and to adopt appropriate legislation. Many other countries in the region are now in the process of developing quarantine guidelines with the view to making them mandatory for movement of aquatic animals and animal products both locally and internationally. It was noted that these attempts have been only partly successful and certain countries find it difficult to implement such guidelines. Considering the regional status of aquatic animal quarantine:

*The consultation **recommended** that an assessment of 'case studies' of successful and unsuccessful attempts at fish quarantine be made part of the process of developing guidelines for governments on quarantine/aquatic animal health legislation. The importance of regional exchange of information on the subject was **recognized***.

4.4.5. In order to produce practical quarantine guidelines, it is imperative that standardized diagnostic methods which can be adopted by the countries of the region be developed. Although such standardized procedures are in place elsewhere, their applicability to the Asian region is questionable. Therefore:

*The consultation **recognized** that the development of high-tech methods for assessing the disease status in Asia would be useful. For example, there are techniques available in Australia for testing for carriers of Aeromonas salmonicida, which might be usefully applied. Polymerase Chain Reaction (PCR) probes and other techniques, including immunological and genetic methods of detecting small numbers of organisms, might also be used in Asia.*

4.4.6. The issue surrounding the trade in aquatic animals and animal products and their disease status will become increasingly linked and important. Certification procedures for fish health are being developed by various agencies and bodies, such as the European Union (EU) and the World Trade Organization (WTO). Considering the recent development of such certification procedures:

*The consultation **recommended** that, in the region, efforts should be made to ensure that any future certification procedures be workable and relevant to the species and the systems of the countries in the region.*

4.4.7. Considering the urgency of developing an effective regional fish health management strategy for Asia and the Pacific: *The consultation **emphasized** the need for a broad assessment of quarantine and fish health legislation, with a view to developing guidelines for assisting governments in policy development. The assessment would involve an analysis of regional experiences and an evaluation of the applicability of existing international codes, including those of OIE, the European Inland Fisheries Advisory Commission (EIFAC), and the International Council for the Exploration of the Sea (ICES). Such an assessment should be carried out through a regional cooperative effort, with the assistance and cooperation of concerned international agencies and bodies, which might include OIE, EIFAC, NACA and FAO. The assessment should cover ornamental fish as well as fish and shellfish produced for human consumption. The importance of involving the industry/private sector in such activities was underlined.*

4.4.8. Since all countries in the region involved in aquaculture production are concerned with the sustainable development of the industry through prevention and control of disease:

*The consultation **agreed** that national aquaculture policies should incorporate relevant aquatic animal health issues. These issues were detailed in the earlier recommendations made by the ADB/NACA Workshop in 1990.*